MXCuBE@NMX

Neutron MX at the European Spallation Source

PRESENTED BY AARON FINKE

5/21/2026

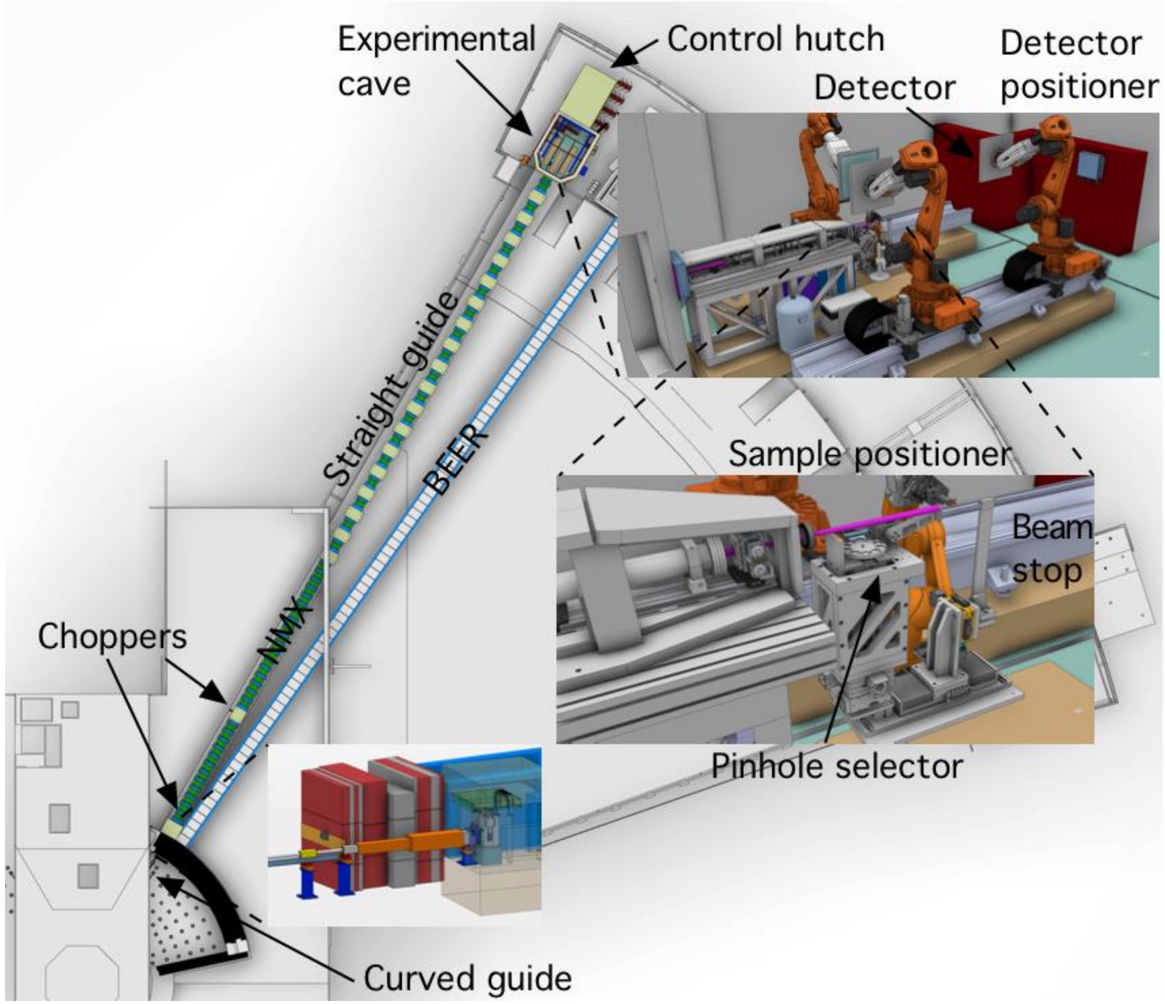
NMX: Neutron Macromolecular Crystallography

A Quasi-Laue TOF instrument

ESS is slated to become the world's most powerful neutron spallation source when it comes online in 2027.

NMX is a quasi-Laue TOF instrument for high-resolution macromolecular crystallography.

Goal: Determination of precise hydrogen atom positions in protein crystals with an innovative detector technology and robot-based goniometry.



The NMX Team

Who's building NMX?

Esko Oksanen
Lead Instrument Scientist

Laís Passine Do Carmo (ECDC)
Controls/MXCuBE

Aaron Finke
Instrument Data Scientist

Zoë Fisher
Deuteration Wizard

Justin Bergmann
Commissioning Instrument Scientist

Swati Aggarwal
Commissioning Scientist

Daniel Lundström
Lead Engineer

ESS and NMX

Current Status

ESS

- Current Beam on Target (first neutrons) date is **Jan 2027**
- Setbacks due to moderator failure from a power glitch (Nov 2025), rotating helium chiller failure (Late 2024)

NMX

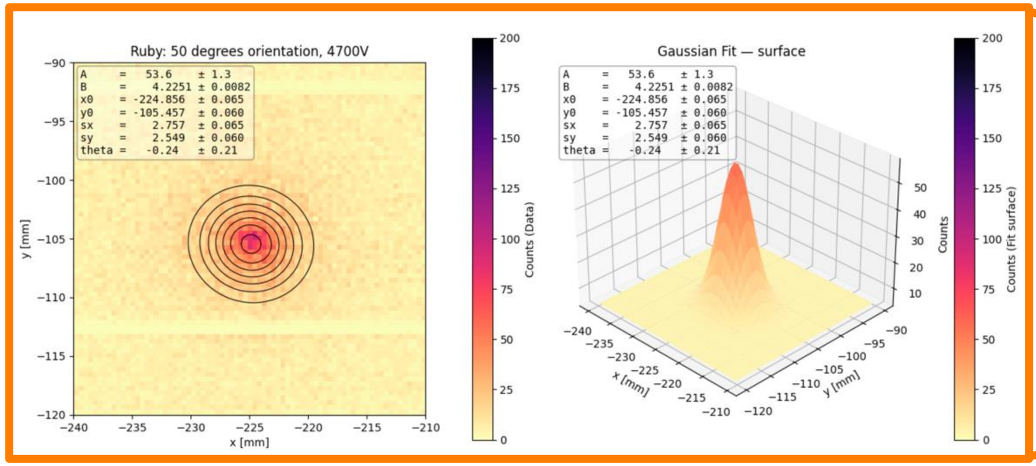
- Taking advantage of delays to test detectors, achieve better readiness for neutrons
- Energization/testing of ARINAX robotic equipment in June
  - overlaps with MXCuBE meeting so we may not be there

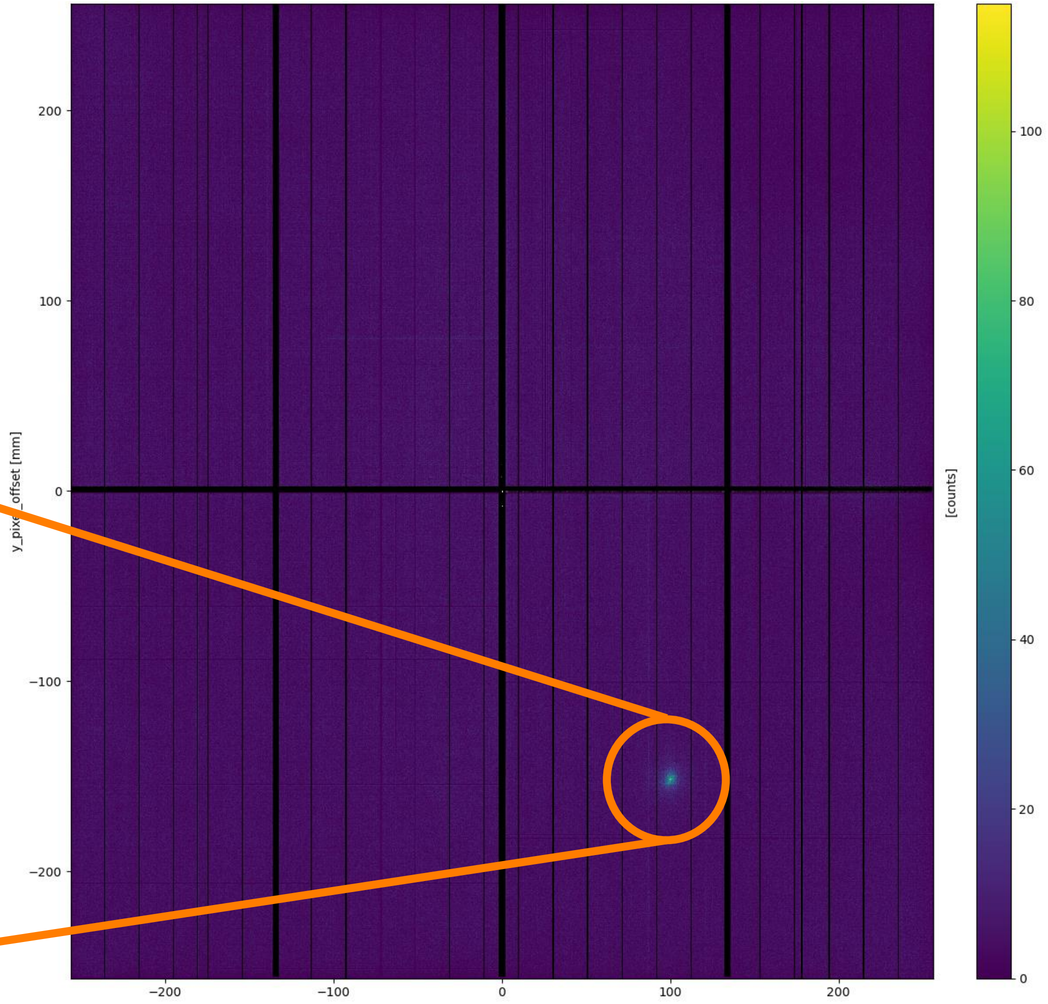
NMX hutch with robot assembly

NMX Update

Testing our Neutron Detectors

- First diffraction on large detector panel (ruby, DALI beamline at ILL)
- Confirmation of successful beam-to-data pipeline with data collections at ILL, BNC
  - Detector -> RMM -> EFU -> Filewriter
  - TOF data not yet tested (maybe at J-PARC?)





Software Updates

MXCuBE

mxcubeweb==4.231.0

mxcubecore==1.188.0

Done

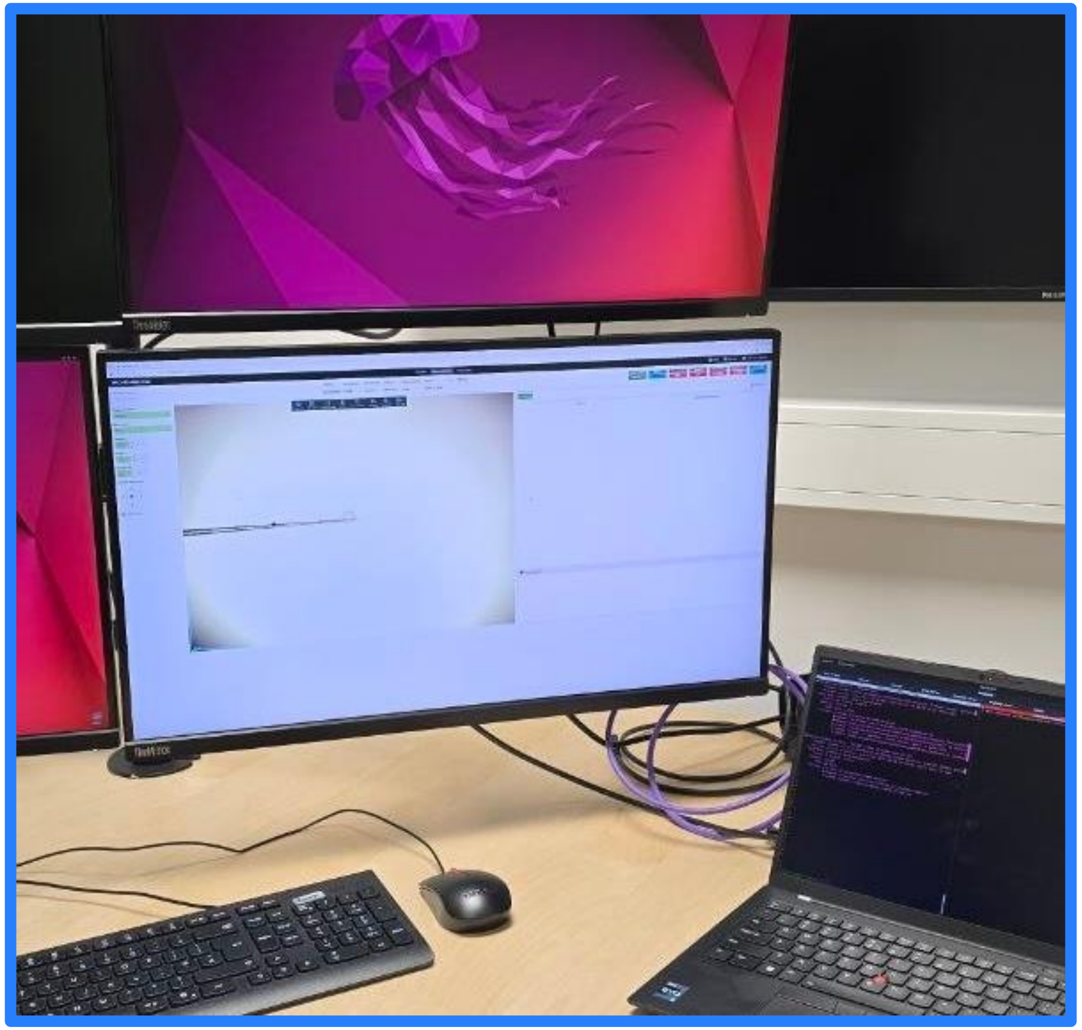
- Base NICOS-MXCuBE communication classes reviewed and merged to mxcubecore (PR #1445)
- MXCuBE deployed to NMX hutch (some read-only controls added)
- ARINAX sample camera image added to MXCuBE in production

Ongoing

- Pre-testing ARINAX EPICS PVs in NICOS/MXCuBE (with the mockup)
- Testing detector configuration (motion) control as a Beamline Action

Next

- On-site test of ARINAX controls and EPICS Pvs
- Pop-up screen for NMX data collection



MXCuBE at NMX hutch (in production) with ARINAX sample image.

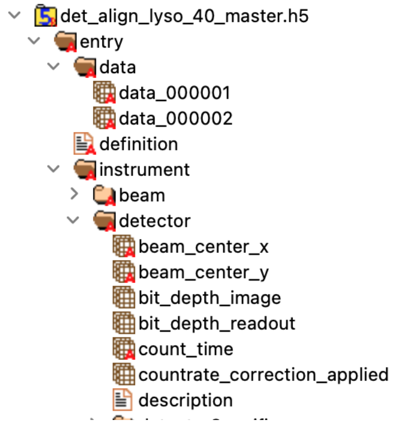
What is SciCat?

The LIMS solution for ESS

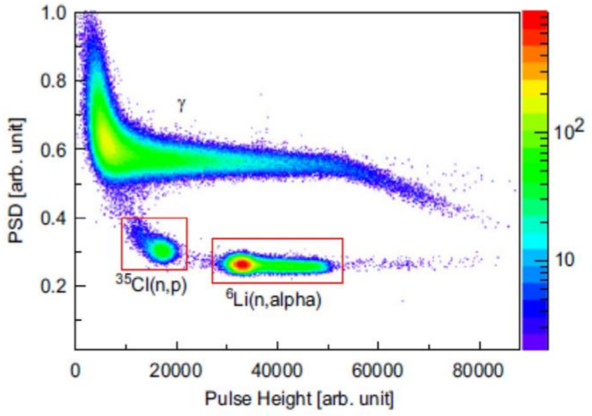
- Metadata catalog management system for scientific data
- Supports management of the entire data lifecycle, from raw to published
- Open source
  - https://scicatproject.github.io/
- Modular design to evolve with the diverse needs of scientists
- Built on modern technical stack
  - Node.js, Angular, Nest.js, MongoDB

Data and Datasets in SciCat

- A “Dataset” is the complete set of data created from an experiment.
  - **Raw Data**: data coming from measurement device in a readable, archivable format
  - **Derived Data**: data calculated from the raw data
- SciCat is a metadata catalog; raw and derived data are stored on the ESS server
- Users can upload electronic and paper notebook pages to SciCat datasets (recommended!)



Raw Data



Derived Data

Dataset Details in SciCat

Datasets / **20.500.12269/88de0000-3edb-4b43-aa53-cc0e76ccb9a6** /

Details | Datafiles | Related Datasets | Logbook | Attachments

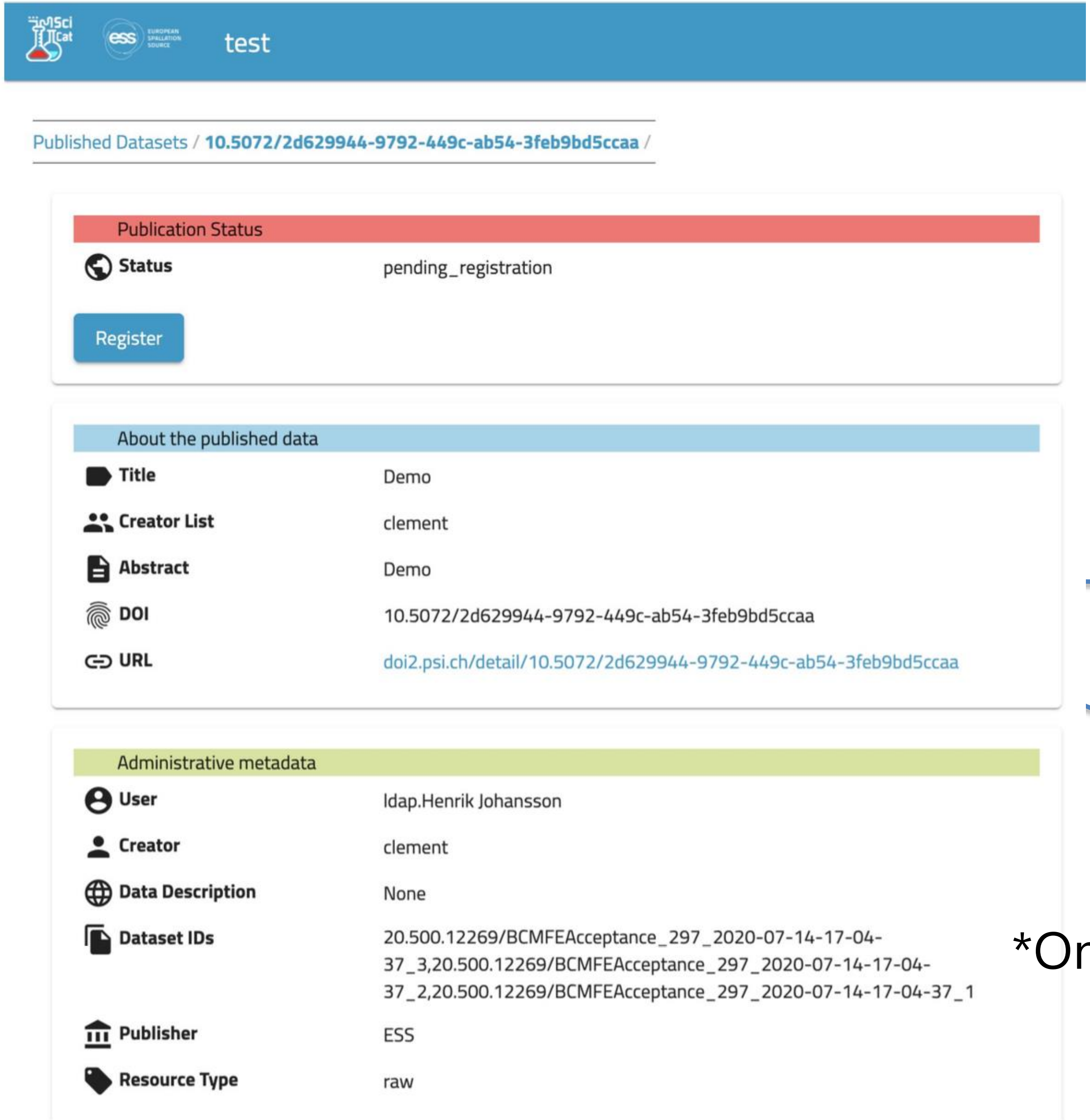
General Information

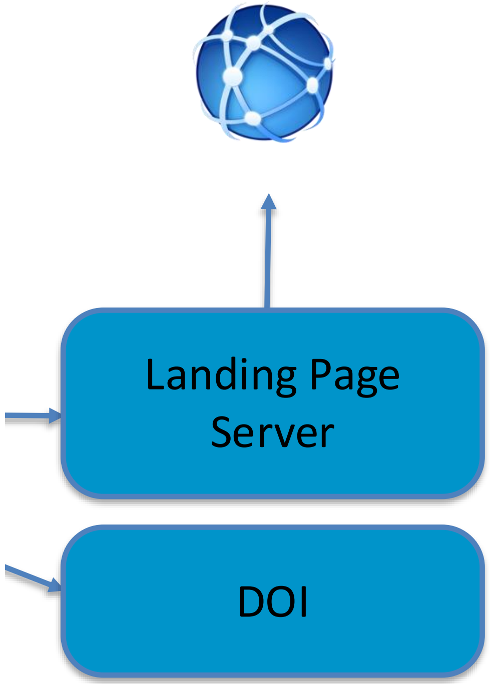
Name	NMX by SCIPP
Description	NMX dataset by SCIPP for DMSC STAP
PID	20.500.12269/88de0000-3edb-4b43-aa53-cc0e76ccb9a6
Type	raw
Creation Time	2024-10-08 18:24
Keywords	NMX, ESS, DMSC_STAP, MCSTAS, SCIPP

Edit

Creator Information

Owner	Aaron Finke
Principal Investigator	Aaron Finke
Contact Email	aaron.finke@ess.eu
Owner Group	668174
Access Groups	ecdc,sims,dram,ess,scientific information management systems group,ess employees





*One DOI can access any number of related datasets

Thank you!